

Learning Curves

A **learning curve** is a diagnostic tool that plots your model's **error** (both J_{train} and J_{cv}) as a function of the **training set size** (m_{train}).

It helps you understand how your algorithm's performance changes as it gets more "experience" (more data).

- **Horizontal Axis:** m_{train} (number of training examples).
 - **Vertical Axis:** Error (e.g., squared error or misclassification error).
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The Shape of a "Good" Learning Curve

For a well-fitting model (one that is "just right"), the learning curves have a predictable shape:

- **Training Error (J_{train}):**
 - When m_{train} is very small (e.g., 1 or 2 examples), the error is **zero or near-zero** because it's easy for the model to perfectly fit a tiny amount of data.
 - As m_{train} **increases**, it becomes harder for the model to perfectly fit every example, so J_{train} **increases** and then flattens out.
- **Cross-Validation Error (J_{cv}):**
 - When m_{train} is very small, the model is trained on tiny, unrepresentative data, so it generalizes terribly. J_{cv} is **very high**.
 - As m_{train} **increases**, the model learns from more data and generalizes better, so J_{cv} **decreases** and then flattens out.

Key Property: For a good model, J_{cv} is typically higher than J_{train} , but the two curves should converge to a low error value as m_{train} gets large.

Using Learning Curves to Diagnose Bias & Variance

Learning curves have very different shapes depending on whether your model has a bias or variance problem.

1. High Bias (Underfitting)

- **What it looks like:** The J_{train} and J_{cv} curves are **both high** and **close to each other**. They flatten out at a high error level.
- **Why it happens:** The model is too simple (like fitting a straight line to a curve). Even with very little data, J_{train} is high because the line can't fit the data. As you add more data, the line doesn't change much, so both errors remain high.

- **Key Takeaway:** If an algorithm has high bias, **getting more training data will not help**. The curves will just stay flat at a high error level.

2. High Variance (Overfitting)

- **What it looks like:** There is a **large gap** between the J_{cv} (which is high) and J_{train} (which is low).
 - **Why it happens:** The model is too complex (like a 10th-order polynomial). It perfectly fits the training data (low J_{train}) but fails to generalize (high J_{cv}).
 - **Key Takeaway:** If an algorithm has high variance, **getting more training data is likely to help**. As m_{train} increases, J_{train} will rise, but J_{cv} will fall, and the gap between them will shrink.
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Practical Use

- Plotting a full learning curve can be computationally expensive (you have to train many models on different subsets of data).
- However, the *concept* is a powerful mental model. The main way to diagnose bias and variance is still to look at the gap between your baseline, J_{train} , and J_{cv} using all your data.